Yong-chan Park

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About me

I am a postdoctoral researcher in Computer Science and Engineering at Seoul National University (SNU), advised by Prof. U Kang in Data Mining Laboratory. My research interests include time-series analysis, tensor analysis, and anomaly detection.

Research vision and current work

My research develops fast, compact, and interpretable algorithms for large-scale data mining, with a focus on frequency analysis, tensor factorization, and latent-factor modeling. I aim to exploit hidden structure in high-dimensional data to improve both computational efficiency and modeling accuracy.

- **Partial Fourier Transform (PFT)** and **Auto-MPFT**: I develop Fourier-based algorithms that accelerate spectral analysis by computing only informative frequency components. This line of work includes PFT [1] for time-series data and Auto-MPFT [5] for multidimensional data with automatic hyperparameter selection.
- Compact tensor factorization via **PuzzleTensor** and **FOCAL**: I study how tensor representations can be transformed or regularized to achieve accurate factorization with lower computational and parameter costs. PuzzleTensor [9] transforms tensor axes to reduce the effective target rank, while FOCAL [14] improves online coupled matrix-tensor factorization through frequency regularization.
- **Masked Mixture Factorization (MMF)**: I develop lightweight extensions of matrix factorization that improve expressiveness without sacrificing interpretability. MMF [15] uses instance-wise masks over a shared latent basis to capture heterogeneous latent structures while retaining the simplicity and scalability of classical factorization methods.

Education

Postdoctoral Researcher — Mar. 2026 – Present
- Computer Science and Engineering, SNU

Ph.D. Student — Mar. 2020 – Feb. 2026
- Computer Science and Engineering, SNU

Bachelor of Science — Mar. 2012 – Feb. 2019
- Department of Mathematical Sciences, SNU

Publications

[1] **Yong-chan Park**, Jun-Gi Jang, and U Kang. “Fast and Accurate Partial Fourier Transform for Time Series Data.” KDD 2021

[2] Jaemin Yoo, Yejun Soun, **Yong-chan Park**, and U Kang. “Accurate Multivariate Stock Movement Prediction via Data-Axis Transformer with Multi-Level Contexts.” KDD 2021

[3] **Yong-chan Park***, Sangjun Son*, Minyong Cho, and U Kang. “DAO-CP: Data-Adaptive Online CP Decomposition for Tensor Stream.” PLOS One 2022 (*equal contribution)

[4] Jun-Gi Jang, Jeongyoung Lee, **Yong-chan Park**, and U Kang. “Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors - Algorithm and Application.” KDD 2023

[5] **Yong-chan Park**, Jongjin Kim, and U Kang. “Fast Multidimensional Partial Fourier Transform with Automatic Hyperparameter Selection.” KDD 2024

[6] Jun-Gi Jang, **Yong-chan Park**, and U Kang. “Fast and Accurate PARAFAC2 Decomposition for Time Range Queries on Irregular Tensors.” CIKM 2024

[7] JinGee Kim, **Yong-chan Park**, Jaemin Hong, and U Kang. “Accurate Stock Movement Prediction via Multi-Scale and Multi-Domain Modeling.” BigData 2024

[8] SeungJoo Lee, **Yong-chan Park**, and U Kang. “Accurate Coupled Tensor Factorization with Knowledge Graph.” BigData 2024

[9] **Yong-chan Park**, Kisoo Kim, and U Kang. “PuzzleTensor: A Method-Agnostic Data Transformation for Compact Tensor Factorization.” KDD 2025

[10] SeungJoo Lee, **Yong-chan Park**, and U Kang. “SwaGNER: Leveraging Span-aware Grid Transformers for Accurate Nested Named Entity Recognition.” CIKM 2025

[11] SeungJoo Lee, **Yong-chan Park**, and U Kang. “Offline and Online Coupled Tensor Factorization with Knowledge Graph.” PLOS One 2025

[12] Nam Kyu Kang, **Yong-chan Park**, and U Kang. “Fast and Accurate Temporal Super-Resolution via Residual-Aware Coupled Tensor Factorization.” ICASSP 2026

[13] SeungJoo Lee, **Yong-chan Park**, and U Kang. “TiRano: Tensorized Relation-aware Temporal Reasoning for Accurate Knowledge Graph Completion.” KDD 2026

[14] **Yong-chan Park**, SeungJoo Lee, and U Kang. “Fast and Accurate Online Coupled Matrix-Tensor Factorization via Frequency Regularization.” KDD 2026

[15] **Yong-chan Park**, SeungJoo Lee, Jeongyoung Lee, and U Kang. “A Masked Mixture Model for Compact and Accurate Matrix Factorization.” KDD 2026